



Booteo de Sistema Operativo Linux

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1. Objetivos

- Aprender a bootear un sistema operativo Linux
- Aprender sobre la secuencia de arranque de un computador como la BIOS y la UEFI

2. Instrucciones

1. Siga el tutorial para la creación de un booteable de Ubuntu como se indica en la unidad 9 en el archivo Arranque Ubuntu¹.
2. Realice un respaldo del código de *BitLocker* utilizando un command prompt de Windows con permisos de administrador²:

```
manage-bde-protectors C:-get
```

3. Utilizando su computador o el del laboratorio, siga los pasos indicados para obtener el booteo desde el USB con el Sistema Operativo de Ubuntu
4. Realice el boot de Ubuntu desde la memoria USB externa y seleccione la opción de *Probar Sistema Operativo*.
5. Una vez ingresado active un terminal de Ubuntu: **Ctrl+ALT+T** y obtenga las características del computador mediante la ejecución de los comandos: **lscpu**, **lsb_release -a** y **lspci**. Obtenga capturas de la ejecución de los comandos. Luego abra este archivo en WSL y ejecute los comandos desde los bloques shell. Para insertar un bloque puede usar la combinación **C-c C-**, escoja la opción **src** e indique que es del tipo shell.

3. Desarrollo Emacs

Se sugiere utilizar la opción **:fontsize scriptsize** para dar formato a la ejecución del comando. Por ejemplo, considere:

```
whoami
```

```
leningfe
```

¹Dependiendo de las características de su computador debe verificar si al crear el medio booteable para Ubuntu, su sistema reconoce o acepta para el *Target System*

²Si no respalda el código, no podrá arrancar el disco luego de reiniciar. Más información en: <https://www.partitionwizard.com/disk-recovery/bitlocker-recovery-key-bypass.html>

3.1. Comando lsb_release

```
lsb_release -a
```

```
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.4 LTS
Release:        22.04
Codename:       jammy
```

3.2. Comando lspci

```
lspci
```

```
64cc:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio 1.0 console (rev 01)
a731:00:00.0 System peripheral: Red Hat, Inc. Virtio file system (rev 01)
e689:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
```

3.3. Comando lscpu

```
lscpu
```

```
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            48 bits physical, 48 bits virtual
Byte Order:              Little Endian
CPU(s):                   2
On-line CPU(s) list:     0,1
Vendor ID:               AuthenticAMD
Model name:              AMD A9-9420 RADEON R5, 5 COMPUTE CORES 2C+3G
CPU family:              21
Model:                   112
Thread(s) per core:      1
Core(s) per socket:      2
Socket(s):               1
Stepping:                0
BogoMIPS:                5988.75
Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
Hypervisor vendor:      Microsoft
Virtualization type:     full
L1d cache:               64 KiB (2 instances)
L1i cache:               192 KiB (2 instances)
L2 cache:                2 MiB (2 instances)
NUMA node(s):            1
NUMA node0 CPU(s):      0,1
Vulnerability Gather data sampling: Not affected
Vulnerability Ghostwrite: Not affected
Vulnerability Indirect target selection: Not affected
Vulnerability Itlb multihit: Not affected
Vulnerability L1tf:      Not affected
Vulnerability Mds:       Not affected
Vulnerability Meltdown:  Not affected
Vulnerability Mmio stale data: Not affected
Vulnerability Old microcode: Not affected
Vulnerability Reg file data sampling: Not affected
```

Vulnerability Retbleed:	Mitigation; untrained return thunk; SMT disabled
Vulnerability Spec rstack overflow:	Not affected
Vulnerability Spec store bypass:	Mitigation; Speculative Store Bypass disabled via prctl
Vulnerability Spectre v1:	Mitigation; usercopy/swapgs barriers and __user pointer sanit
Vulnerability Spectre v2:	Mitigation; Retpolines; IBPB conditional; STIBP disabled; RSE
Vulnerability Srbds:	Not affected
Vulnerability Tsa:	Not affected
Vulnerability Tsx async abort:	Not affected
Vulnerability Vmscape:	Not affected

4. Desarrollo Capturas

Coloque las capturas de los resultados obtenidos en los terminales.

```

ubuntu@ubuntu:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                2
On-line CPU(s) list:   0,1
Vendor ID:              AuthenticAMD
Model name:             AMD A9-9420 RADEON R3, 5 COMPUTE CORES 2C+3G
CPU family:            21
Model:                 112
Thread(s) per core:    1
Core(s) per socket:    2
Socket(s):              1
Stepping:               0
Frequency boost:        enabled
CPU(s) scaling MHz:    90%
CPU max MHz:           3000.0000
CPU min MHz:           1400.0000
BogoMIPS:               5989.07
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pg
                        e mca cmov pat pse36 clflush mmx fxsr sse sse2 ht s
                        yscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constan
                        t_tsc rep_good acc_power nopl nonstop_tsc cpuid ext
                        d_apicid aperfmperf pni pclmulqdq monitor ssse3 fma
                        cx16 sse4_1 sse4_2 novbe popcnt aes xsave avx f16c
                        lahf_lm cpl legacy svm extapic cr8_legacy abm sse4
                        a misalmsse 3dnowprefetch osvw ibs xop skinit wdt
                        lwp fma4 tce nodeid msr tbn perfctr_core perfctr_n
                        b bpext ptsc mwaitx cpb hw_pstate ssbd vmcall fsgs
                        base bmi1 avx2 smep bmi2 xsaveopt arat npt lbrv svm
                        _lock nrip save tsc scale vncb_clean flushbyasid de
                        codeassists pausefilter pfthreshold avic v_vmsave_v
                        mload vgif overflow_recov
Virtualization features:
Virtualization:         AMD-V
Caches (sum of all):
L1d:                    64 KiB (2 instances)

```

Figura 1: Captura del comando lscpu - parte 1

```
Jul 10 04:47
ubuntu@ubuntu: ~
base bmi1 avx2 smep bmi2 xsaveopt arat npt lbrv svm
_lock nrip save tsc scale vncb clean flushbyasid de
codeassist pausefilter pfthreshold avic v_vmsave_v
mload vgif overflow_recov

Virtualization features:
Virtualization: AMD-V
Caches (sum of all):
L1d: 64 KiB (2 instances)
L1i: 192 KiB (2 instances)
L2: 2 MiB (2 instances)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0,1
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Not affected
Retbleed: Mitigation; untrained return thunk; SMT disabled
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via p
rtl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user poi
nter sanitization
Spectre v2: Mitigation; Retpolines; STIBP disabled; RSB filling
; PBRSE-eIBRS Not affected; BHI Not affected
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
ubuntu@ubuntu:~$
```

Figura 2: Captura del comando lscpu - parte 2

```
ubuntu@ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 26.04 LTS
Release: 26.04
Codename: resolute
ubuntu@ubuntu:~$
```

Figura 3: Captura del comando lsb_release

```
ubuntu@ubuntu:~$ lspci
00:00.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) Processor Root Complex
00:00.2 IOMMU: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) I/O Memory Management Unit
00:01.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Stoney [Radeon R2/R3/R4/R5 Graphics] (rev da)
00:01.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Stoney HDMI/DP Audio Controller
00:02.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) Host Bridge
00:02.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) Processor Root Port
00:02.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) Processor Root Port
00:02.4 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) Processor Root Port
00:03.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) Host Bridge
00:08.0 Encryption controller: Advanced Micro Devices, Inc. [AMD] Carrizo Platform Security Processor
00:09.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Carrizo Audio Dummy Host Bridge
00:09.2 Audio device: Advanced Micro Devices, Inc. [AMD] Family 15h (Models 60h-6fh) Audio Controller
00:10.0 USB controller: Advanced Micro Devices, Inc. [AMD] FCH USB XHCI Controller (rev 20)
00:11.0 SATA controller: Advanced Micro Devices, Inc. [AMD] FCH SATA Controller [AHCI mode] (rev 4b)
00:12.0 USB controller: Advanced Micro Devices, Inc. [AMD] FCH USB EHCI Controller (rev 49)
00:14.0 SMBus: Advanced Micro Devices, Inc. [AMD] FCH SMBus Controller (rev 4b)
00:14.3 ISA bridge: Advanced Micro Devices, Inc. [AMD] FCH LPC Bridge (rev 11)
00:18.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Stoney HT Configuration
00:18.1 Host bridge: Advanced Micro Devices, Inc. [AMD] Stoney Address Maps
00:18.2 Host bridge: Advanced Micro Devices, Inc. [AMD] Stoney DRAM Configuration
00:18.3 Host bridge: Advanced Micro Devices, Inc. [AMD] Stoney Miscellaneous Configuration
00:18.4 Host bridge: Advanced Micro Devices, Inc. [AMD] Stoney PM Configuration
00:18.5 Host bridge: Advanced Micro Devices, Inc. [AMD] Stoney NB Performance Monitor
01:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Sun XT [Radeon HD 8670A/8670M/8690M / R5 M330 / M430 / Radeon 520 Mobile] (rev 83)
02:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168/8211/8411 PCI Express Gigabit Ethernet Controller (rev 15)
03:00.0 Network controller: Realtek Semiconductor Co., Ltd. RTL8723BE PCIe Wireless Network Adapter
ubuntu@ubuntu:~$
```

Figura 4: Captura del comando lspci

5. Consulta

Describe que hacen y para qué sirven los comandos utilizados en la práctica

lscpu Muestra la información detallada sobre la arquitectura del procesador. Sirve para identificar características claves del hardware, fabricante, modelo, cantidad de núcleos y memoria caché, es muy útil para diagnosticar las capacidades del sistema.

lsb\release -a Muestra información específica sobre la distribución de Linux que está instalada. El parámetro '-a' indica que imprima todos los detalles posibles.

lspci Es una herramienta que lista todos los buses PCI del computador y los dispositivos de hardware que están conectados a ellos. Sirve para identificar componentes físicos internos de la máquina, es fundamental para tareas de diagnóstico de hardware y búsqueda de controladores o drivers.